5-MINUTE REVISION

Factorial Experiments

From the chapter's own revision points, formulae and questions.

Selected from Chapter 21 · Chapter Summary in the textbook.

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STAT 3202 · Chapter revision

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Quick revision · 1

- Factorial experiment → studies two or more factors simultaneously.
- Treatment combinations → all possible combinations of factor levels.
- Number of treatments → product of the number of levels of all factors.
- Main effect → average change in response due to a change in the level of one factor.
- Simple effect → effect of one factor at a particular level of another factor.

Quick revision · 2

- Interaction → effect of one factor changes according to the level of another factor.
- Parallel population mean profiles → no interaction on the response scale used.
- Non-parallel sample mean profiles → an interaction pattern; an appropriate test is needed to assess statistical significance.
- Symmetrical factorial → all factors have the same number of levels.
- Asymmetrical factorial → factors have unequal numbers of levels.

Quick revision · 3

- Complete factorial → all possible treatment combinations are included.
- Incomplete or fractional factorial → only part of the possible combinations are included.
- 2^n factorial → n factors, each at two levels.
- 3^n factorial → n factors, each at three levels.
- 2^2 → 4 treatment combinations.

Quick revision · 4

- 2^3 → 8 treatment combinations.
- 2^4 → 16 treatment combinations.
- 3^2 → 9 treatment combinations.
- 3^3 → 27 treatment combinations.
- In a 2^n factorial, the treatment degrees of freedom are $2^n - 1$.

Quick revision · 5

- In a 3^n factorial, the treatment degrees of freedom are $3^n - 1$.
- In a 2^2 factorial → A, B, and $A \times B$ each have 1 degree of freedom.
- In a 2^3 factorial → A, B, C, $A \times B$, $A \times C$, $B \times C$, and $A \times B \times C$ each have 1 degree of freedom.
- In a 3^2 factorial → A has 2 df, B has 2 df, and $A \times B$ has 4 df.
- For a three-level factor → 2 df can be divided into linear and quadratic components.

Quick revision · 6

- Yates's method → tabular method for calculating effects and sums of squares in 2^n factorial experiments.
- Fisher's algebraic method → expresses factorial effects and interactions algebraically.
- Confounding → deliberately making an interaction inseparable from block effects to reduce block size.
- Complete confounding → the same interaction is confounded in all replications.
- Partial confounding → different interactions are confounded in different replications.

Quick revision · 7

- Higher-order interactions are generally preferred for confounding because main effects and lower-order interactions are usually more important.
- Major advantage of confounding → smaller and more homogeneous blocks, resulting in improved precision.
- Major disadvantage of confounding → loss of information about the confounded interactions.

Important formulae · 1

Number of treatment combinations in a factorial experiment:

$$N = \prod_{i=1}^{n} s_i$$

{#eq-ntreat}

Number of treatment combinations in a 2^n factorial experiment:

$$N = 2^n$$

{#eq-n2n}

Number of treatment combinations in a 3^n factorial experiment:

$$N = 3^n$$

{#eq-n3n}

Important formulae · 2

Main effect of A in a 2^2 factorial experiment:

$$\text{Main effect of A} = \frac{(A)_{i=1} - (A)_{i=0}}{2r}$$

{#eq-maineffA22}

Sum of squares due to A in a 2^2 factorial experiment:

$$SS_A = \frac{[(A)_{i=1} - (A)_{i=0}]^2}{4r}$$

{#eq-ssA22}

Main effect of B in a 2^2 factorial experiment:

$$\text{Main effect of B} = \frac{(B)_{j=1} - (B)_{j=0}}{2r}$$

{#eq-maineffB22}

Important formulae · 3

Sum of squares due to B in a 2^2 factorial experiment:

$$SS_B = \frac{[(B)_{j=1} - (B)_{j=0}]^2}{4r}$$

{#eq-ssB22}

Interaction effect of $A \times B$ in a 2^2 factorial experiment:

$$\text{Interaction effect of } A \times B = \frac{(AB)_{i+j=0} - (AB)_{i+j=1}}{2r}$$

{#eq-intAB22}

Sum of squares due to $A \times B$:

$$SS_{AB} = \frac{[(AB)_{i+j=0} - (AB)_{i+j=1}]^2}{4r}$$

{#eq-ssAB22}

Important formulae · 4

Critical difference for a significant interaction in a 2^2 factorial experiment:

$$CD = t_{\alpha/2,edf}\sqrt{\frac{2MSE}{r}}$$

{#eq-cd22int}

Main effect of A in a 2^3 factorial experiment:

$$\text{Main effect of A} = \frac{(A)_{i=1} - (A)_{i=0}}{4r}$$

{#eq-maineffA23}

Sum of squares due to A in a 2^3 factorial experiment:

$$SS_A = \frac{[(A)_{i=1} - (A)_{i=0}]^2}{8r}$$

{#eq-ssA23}

Important formulae · 5

Interaction effect of $B \times C$ in a 2^3 factorial experiment:

$$\text{Interaction effect of } B \times C = \frac{(BC)_{j+k=0} - (BC)_{j+k=1}}{4r}$$

{#eq-intBC23}

Sum of squares due to $B \times C$:

$$SS_{BC} = \frac{[(BC)_{j+k=0} - (BC)_{j+k=1}]^2}{8r}$$

{#eq-ssBC23}

Interaction effect of $A \times B \times C$:

$$\text{Interaction effect of } A \times B \times C = \frac{(ABC)_{i+j+k=1} - (ABC)_{i+j+k=0}}{4r}$$

{#eq-intABC}

Important formulae · 6

Sum of squares due to $A \times B \times C$:

$$SS_{ABC} = \frac{[(ABC)_{i+j+k=1} - (ABC)_{i+j+k=0}]^2}{8r}$$

{#eq-ssABC}

Critical difference for comparing means of a two-factor interaction in a 2^3 factorial:

$$CD = t_{\alpha/2,edf}\sqrt{\frac{MSE}{r}}$$

{#eq-cd23twofactor}

Critical difference for comparing means when the three-factor interaction is significant:

$$CD = t_{\alpha/2,edf}\sqrt{\frac{2MSE}{r}}$$

{#eq-cd23threefactor}

Important formulae · 7

Linear comparison for a factor at three levels:

$$A_L = (A)_{i=2} - (A)_{i=0}$$

{#eq-linearcomp}

Quadratic comparison for a factor at three levels:

$$A_Q = (A)_{i=2} + (A)_{i=0} - 2(A)_{i=1}$$

{#eq-quadcomp}

Sum of squares due to A in a 3^2 factorial:

$$SS_A = \frac{(A)_{i=0}^2 + (A)_{i=1}^2 + (A)_{i=2}^2}{3r} - CF$$

{#eq-ssA32}

Important formulae · 8

Sum of squares due to B in a 3^2 factorial:

$$SS_B = \frac{(B)_{j=0}^2 + (B)_{j=1}^2 + (B)_{j=2}^2}{3r} - CF$$

{#eq-ssB32}

Sum of squares due to $A \times B$ in a 3^2 factorial:

$$SS_{AB} = SS_{\text{two-way table}} - (SS_A + SS_B)$$

{#eq-ssAB32}

Critical difference for comparing levels of A in a 3^2 factorial:

$$CD = t_{\alpha/2,edf}\sqrt{\frac{2MSE}{3r}}$$

{#eq-cdA32}

Important formulae · 9

Critical difference for comparing levels of A in a 2×3 factorial:

$$CD_A = t_{\alpha/2,edf}\sqrt{\frac{2MSE}{3r}}$$

{#eq-cdA23asym}

Critical difference for comparing levels of B in a 2×3 factorial:

$$CD_B = t_{\alpha/2,edf}\sqrt{\frac{2MSE}{2r}}$$

{#eq-cdB23asym}

Critical difference for comparing means when $A \times B$ is significant:

$$CD = t_{\alpha/2,edf}\sqrt{\frac{2MSE}{r}}$$

{#eq-cdAB23asym}

Self-check from the chapter

1. An experiment containing all combinations of the levels of its factors is called a _____ factorial experiment. **[Full]**
2. The number of treatment combinations is the _____ of the numbers of levels of the factors. **[Product]**
3. The effect of a factor averaged over the levels of the other factors is called its _____ effect. **[Main]**
4. The effect of one factor at a specified level of another is called a _____ effect. **[Simple]**